

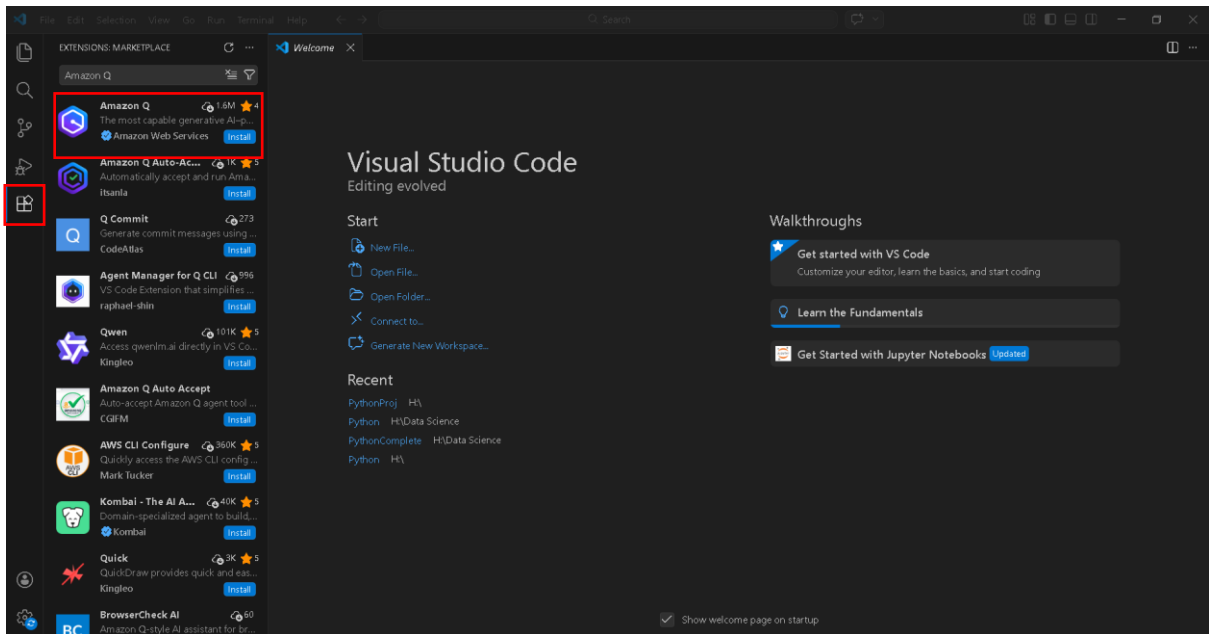
AMAZON Q DEVELOPER

Amazon Q Developer: -

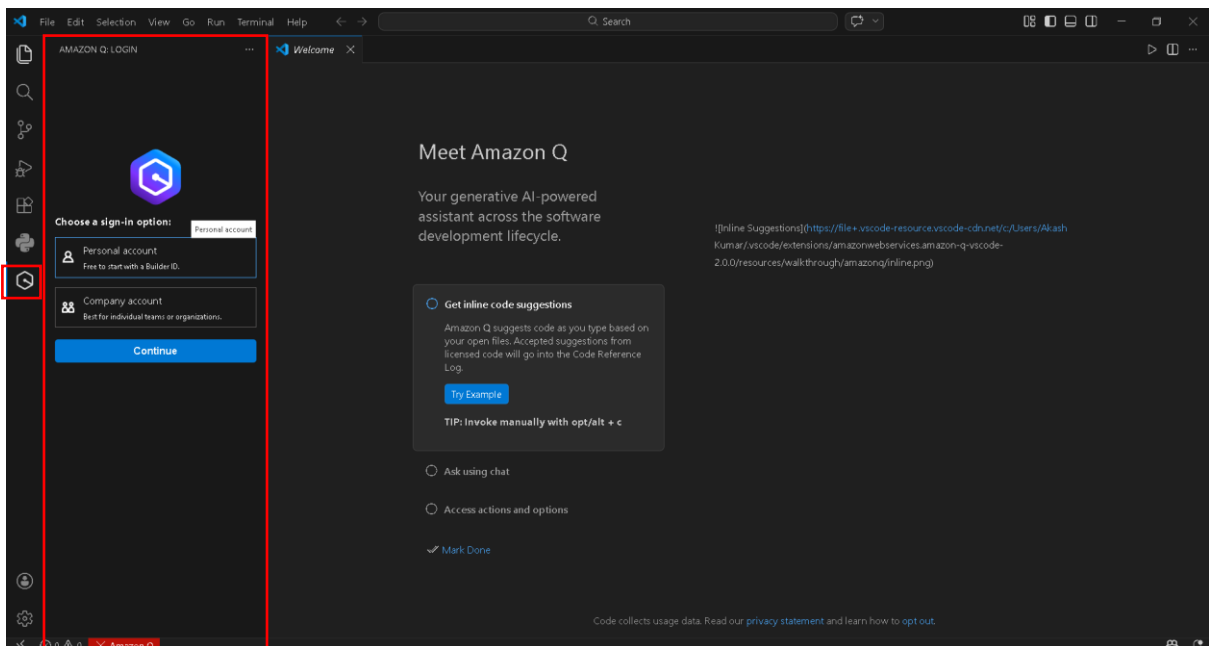
- Amazon Q Developer is an AI-powered assistant that helps developers build, analyze, and manage applications on AWS by generating code, answering questions, and improving software quality.
- It can Assist in coding and testing.
- It supports all the major programming language like java, python, Go etc.
- It helps in diagnosing errors
- It can also perform security scanning and fixes.
- Where to Access Amazon Q – Developer from: -
 - AWS Management Console
 - AWS CLI
 - IDE such as VS code

How Amazon Q can helps in Software Development Lifecycle: -

- **Planning: -**
 - Business Requirement Document(Epic) : -
 - Ex: - A store inventory manager, I should be able to check the inventory details for items in the store and update them when new items are received at the store.
 - Features and User Stories: -
 - Based on Epic we create our Features and User Stories.
- For this phase we can use Amazon Q Business
- In Amazon Q Business Jira Plugin and you can write the requirements in the Natural language in the chat interface.
- And then it creates Features and User stories automatically in the JIRA.
- **Design: -**
 - In the Design Phase Amazon Q Developer can help you generate Architecture for the use case.
 - So now Let's see how: -
 - The first thing you need to download VS Code.



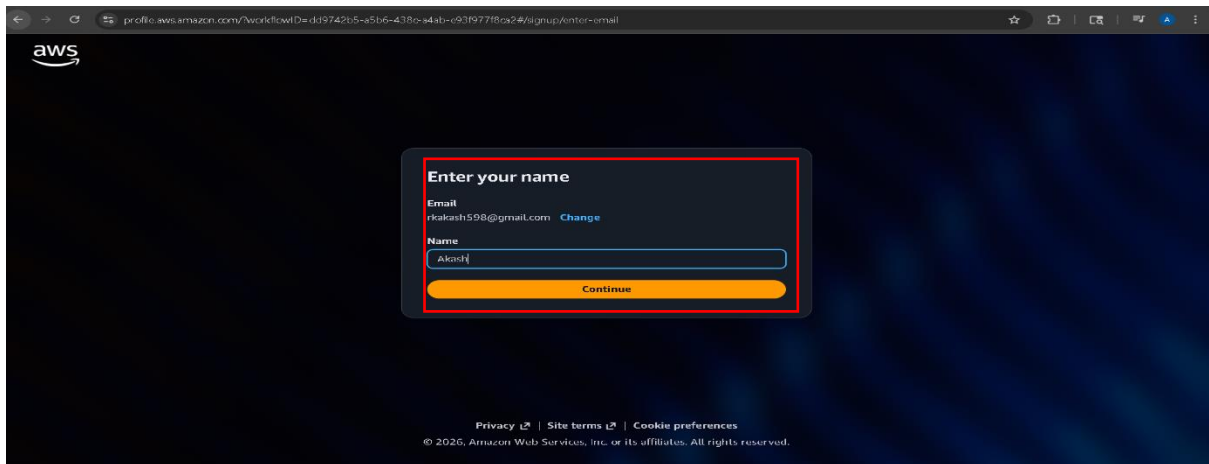
- Open it and click on Extension icon (left side Fifth icon)
- And Search Amazon Q
- Click on it and install it



- Click on sign in with personal account.
- But for sign in you need a Builder id

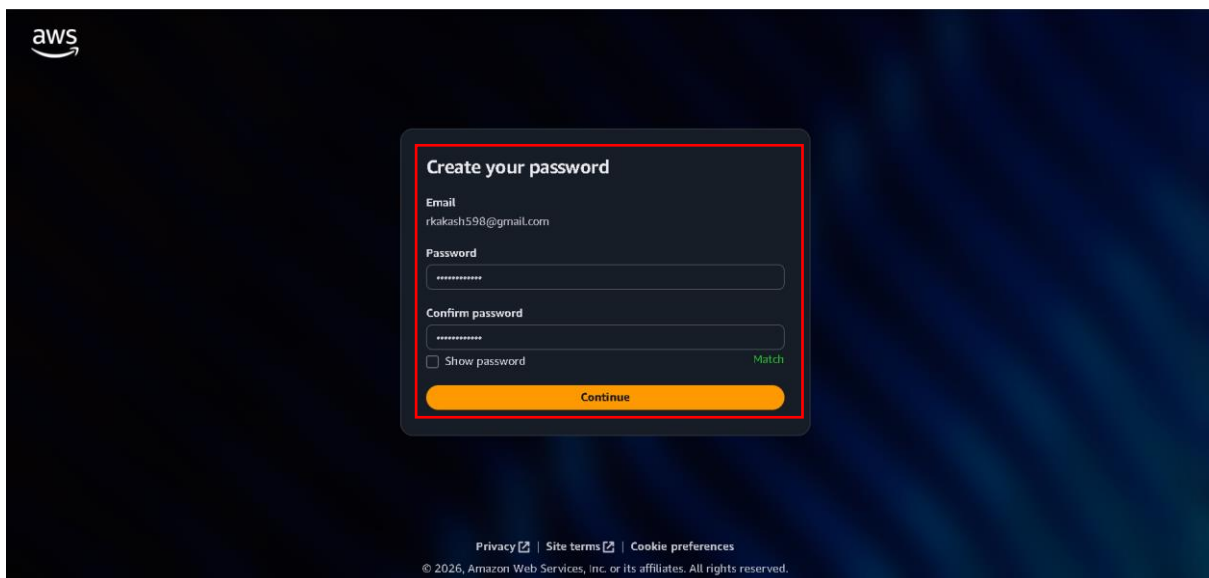
So let's create a Builder id: -

- Open any Browser and search AWS Builder ID.



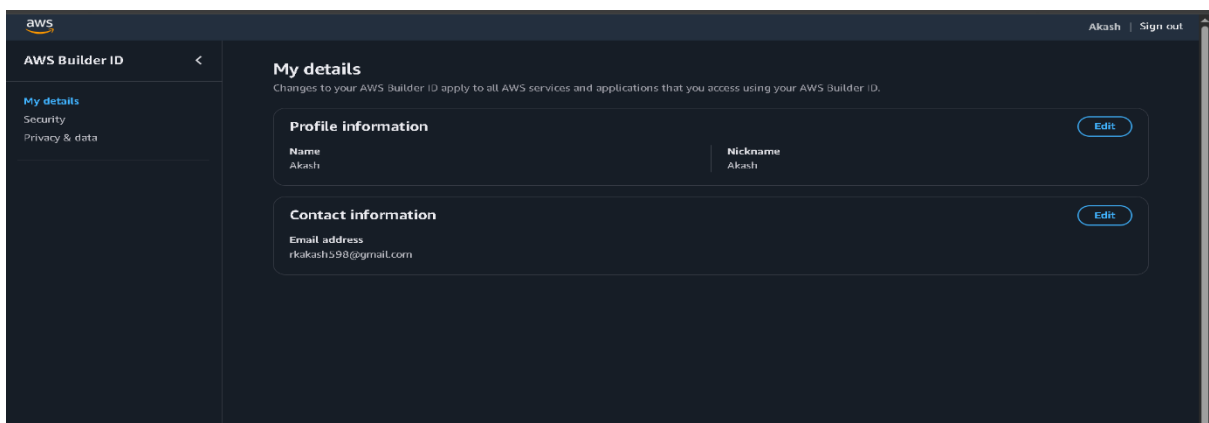
The screenshot shows the AWS Builder ID registration page. The form is titled "Enter your name" and is highlighted with a red border. It contains the following fields: "Email" (pre-filled with "rkakash598@gmail.com" and a "Change" link), "Name" (pre-filled with "Akash"), and a "Continue" button. At the bottom, there are links for "Privacy", "Site terms", and "Cookie preferences", and a copyright notice for 2026 Amazon Web Services.

- Enter Email and Name and click continue
- Enter verification code



The screenshot shows the "Create your password" form, highlighted with a red border. It includes the "Email" field (pre-filled with "rkakash598@gmail.com"), "Password" and "Confirm password" fields (both masked with dots), a "Show password" checkbox, and a "Continue" button. A green "Match" indicator is visible next to the confirm password field. The footer contains the same links and copyright notice as the previous screen.

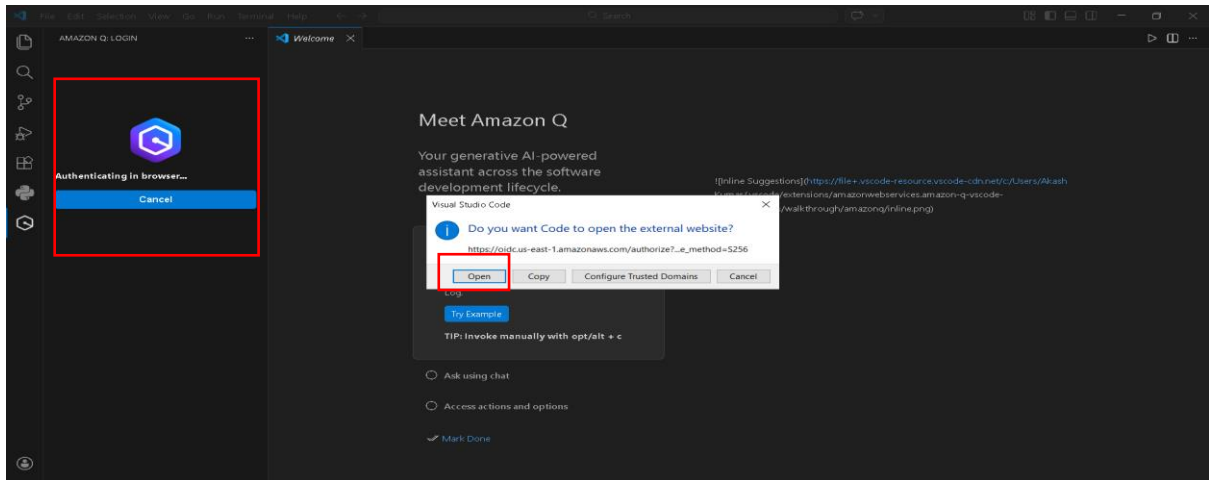
- Now Enter Password



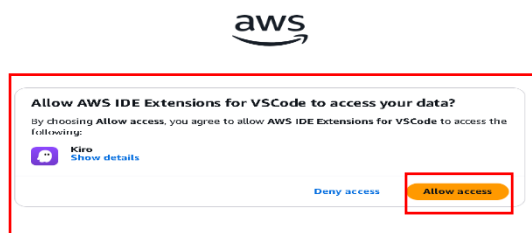
The screenshot shows the "My details" page of the AWS Builder ID console. The left sidebar has a "My details" link. The main content area is titled "My details" and includes a subtitle: "Changes to your AWS Builder ID apply to all AWS services and applications that you access using your AWS Builder ID." There are two sections: "Profile information" (with fields for "Name" and "Nickname", both pre-filled with "Akash", and an "Edit" button) and "Contact information" (with a field for "Email address" pre-filled with "rkakash598@gmail.com" and an "Edit" button).

- Now your Builder id is Ready

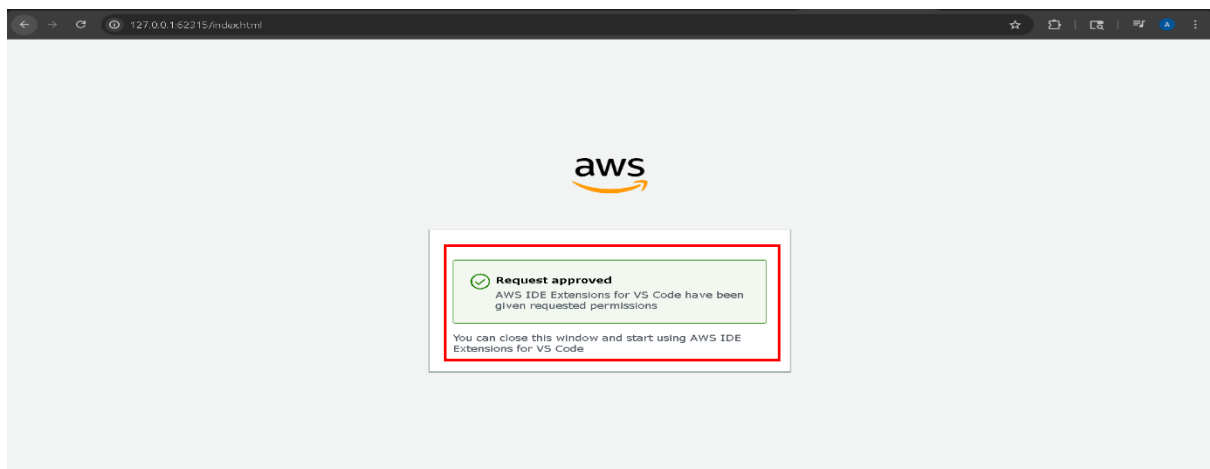
- Now Back to your Vs Code and go in Amazon Q
- Choose Sign in personal account
- Click continue.



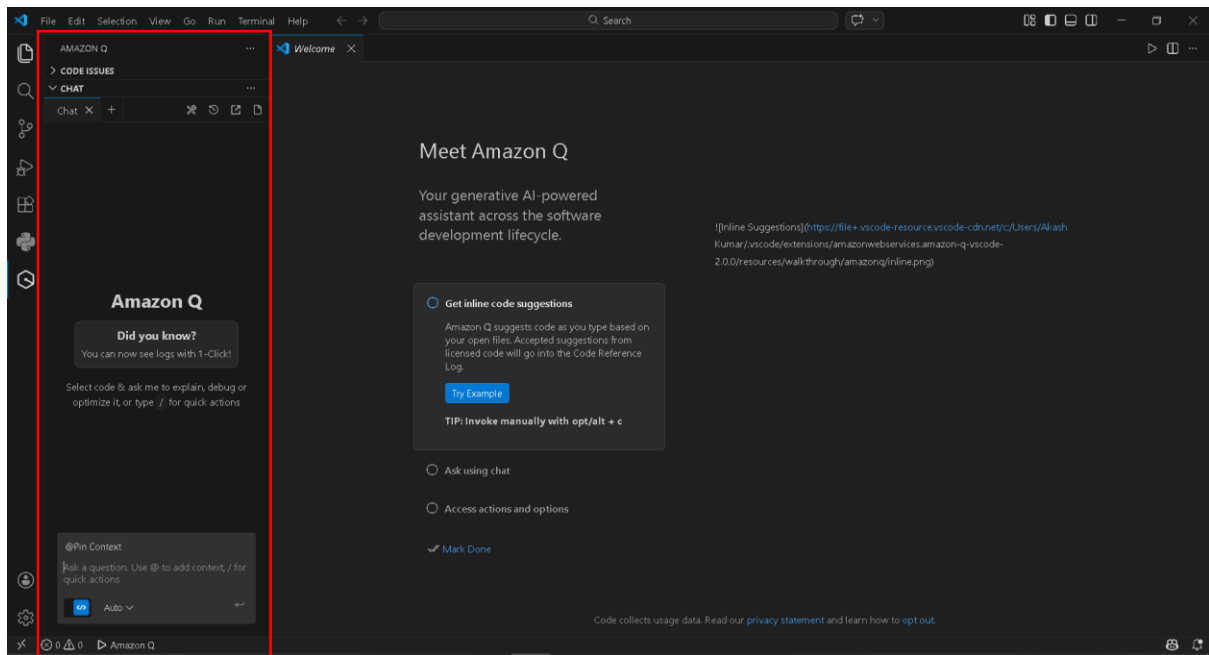
- Now click on Open
- Redirected to your browser



- Now Click Allow Access

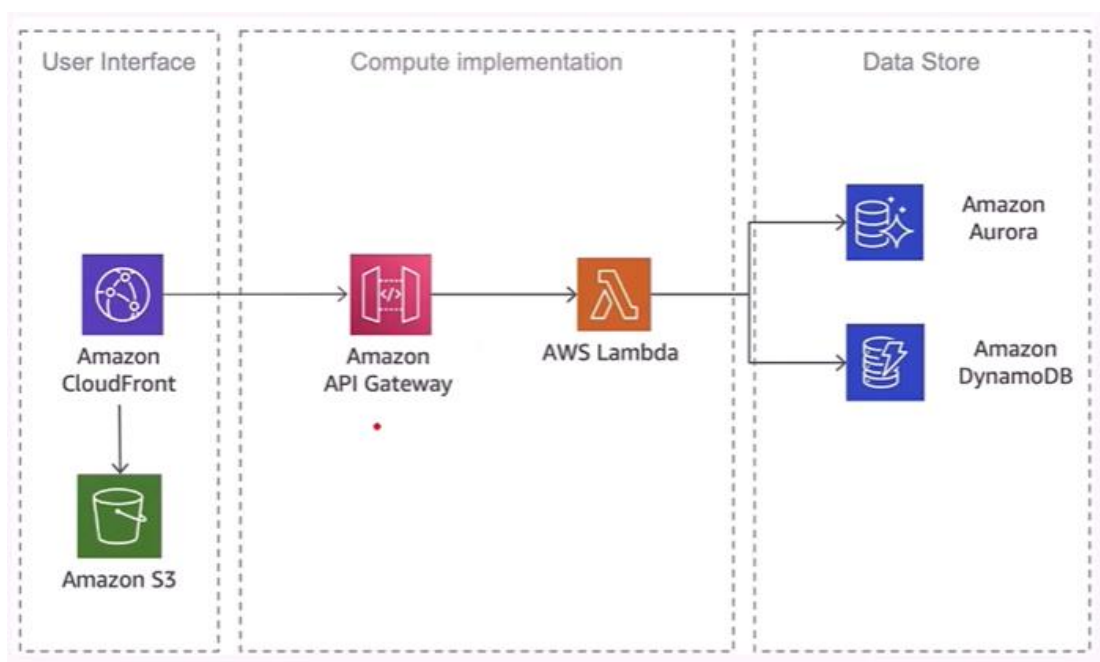


- Your request is approved



- Now our Amazon Q is ready to ask any question; it's a LLM

Now Understand about AWS Micro-Services Architecture: -



• Now this is our Flow: -

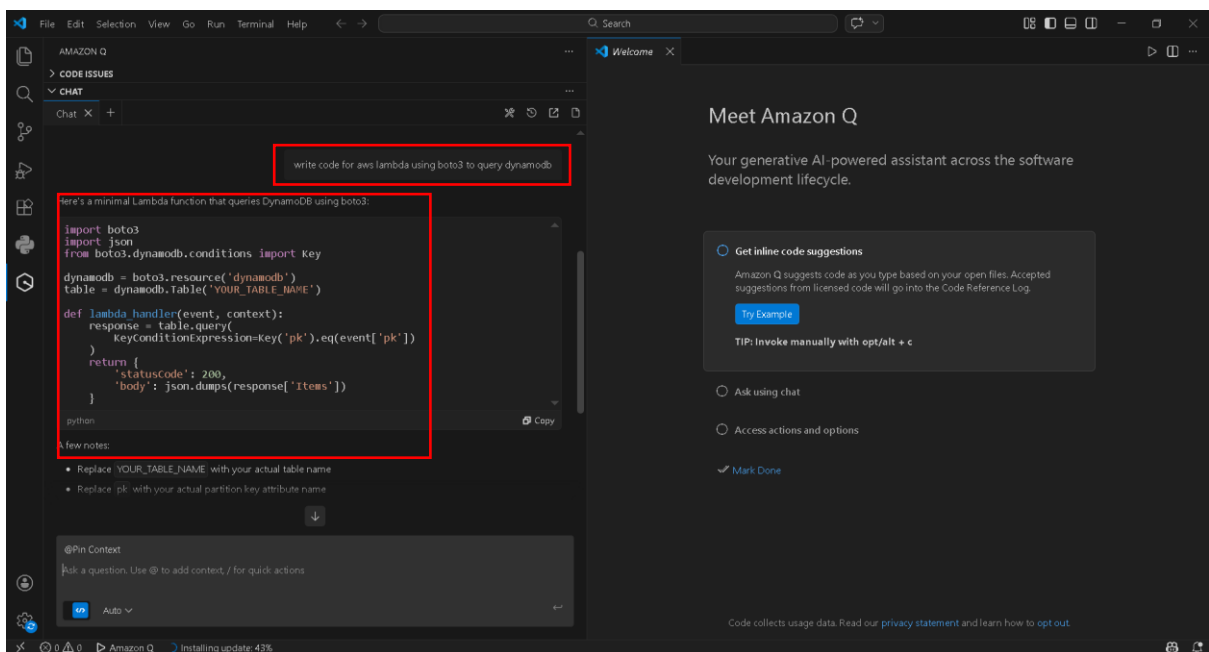
- We will send the request to Amazon API gateway
- Which invokes the Amazon Lambda Function
- And then Amazon Lambda makes a query to the Amazon Dynamo DB

- **Implementation: -**

- Coding
- Debugging
- Improve code
- Unit Testing

- **Example : -Now let's generate the code for this Architecture: -**

- Now let's we start coding
- In which we'll have write the Lambda function which will query the Amazon Dynamo DB
- And we are going to use python SDK Boto3 for doing this.
- So let's Ask Amazon Q Developer to Generate the code



- Now you see it write code for us.
- Also explain and fix your already written code.

- **Testing: -**

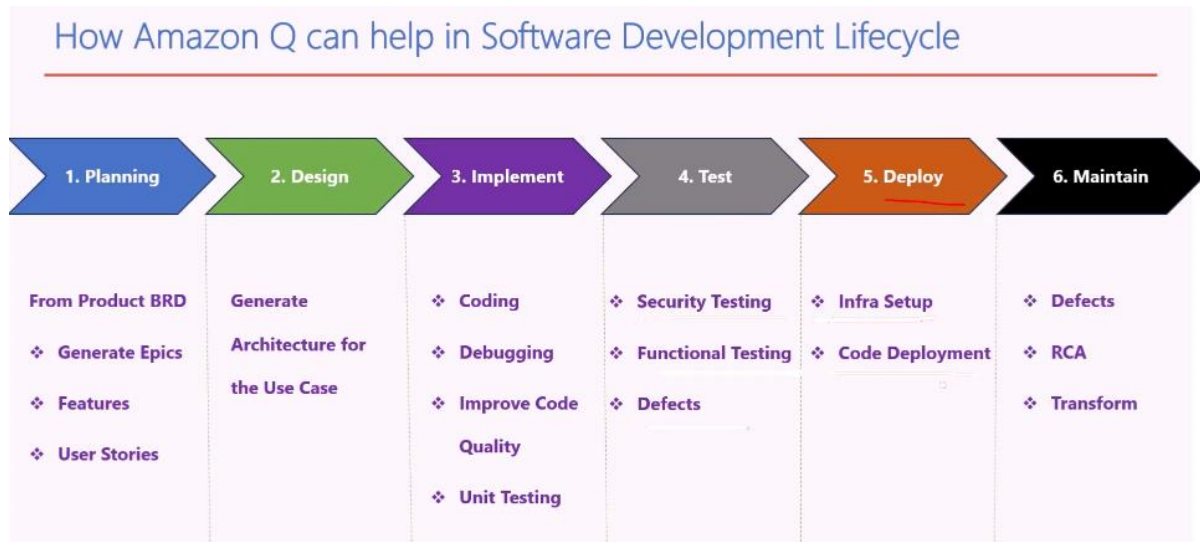
- Security Testing
- Functional Testing
- Defects

- **Deploy: -**

- Infra setup
- Code Deployment

- **Maintain: -**

- Defect
- RCA
- Transform



NOTE: - This Note to give you a overview of how Amazon Q Developer can help across software development.